



BITS Pilani

Hyderabad Campus

Analog Electronics

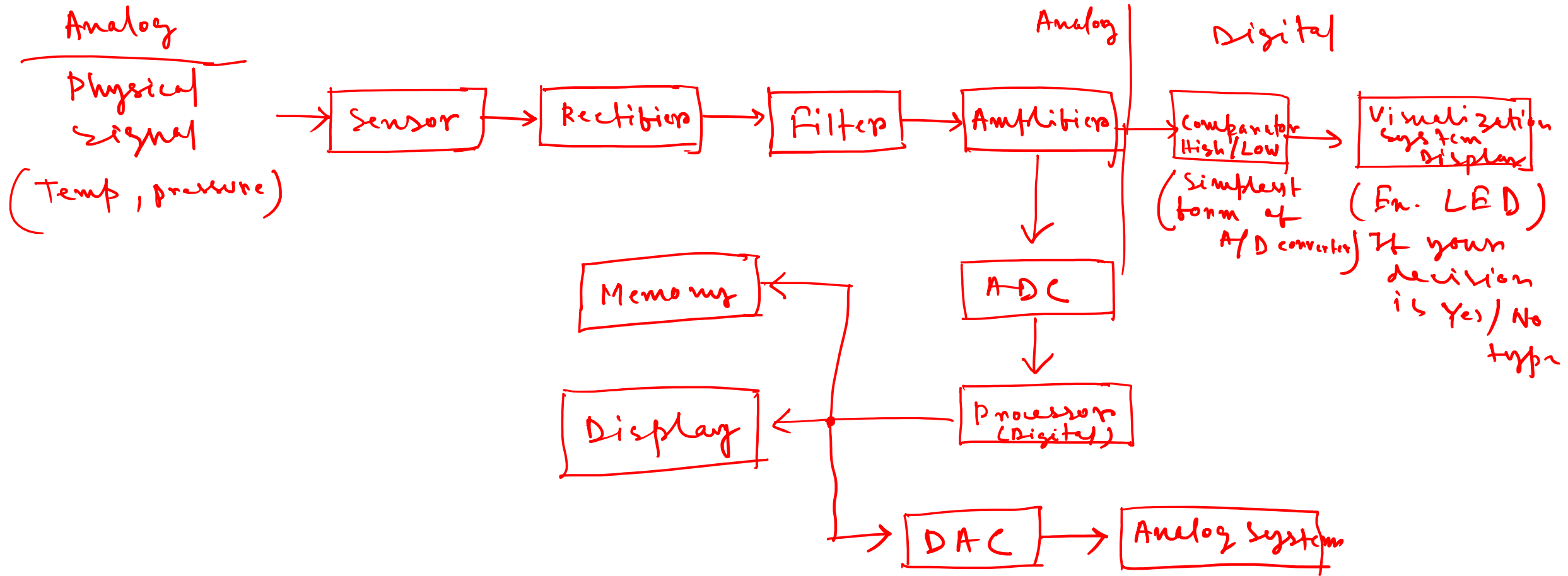
Lecture-36 and 37: DAC

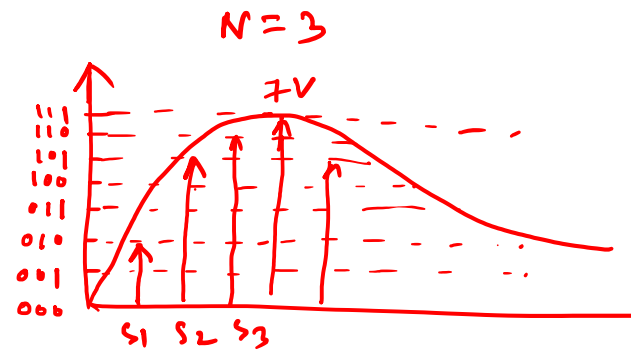
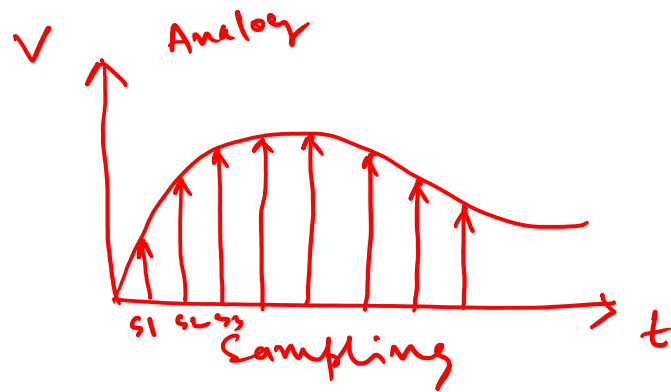
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DAC and ADC





$$s_1 \Rightarrow 010$$

$$s_2 \Rightarrow 101$$

Quantization
 (Approximation process
 to choose the sample
 amplitude closer
 to the one of the
 quantization level)

MSB			LSB			Decimal Equivalent	Corresponding Voltage	
b_2	b_1	b_0						
0	0	0				0	0V	0V
0	0	1				1	1V	0.1V
0	1	0				2	2V	0.2V
0	1	1				3	3V	0.3V
1	0	0				4	4V	0.4V
1	0	1				5	5V	0.5V
1	1	0				6	6V	0.6V
1	1	1				7	7V	0.7V

Resolution $\Rightarrow$ Change of
 voltage due to 1b change
 of LSB.

$$\text{Resolution} = 1V$$

$$\Delta V = 0.1V \text{ (Resolution)}$$

Design a digital voltmeter having
 resolution 0.1V in the range 0 to 1V.
 How many bits required

$$V_{FS} = 1, \Delta V = 0.1$$

$$\text{No of states} = \frac{V_{FS}}{\Delta V} = \frac{1}{0.1} = 10$$

$$2^N \geq 10, \quad \underline{N = 4}$$

$$\text{For } \Delta V = \underline{0.01V}, \quad \frac{V_{FS}}{\Delta V} = 100$$

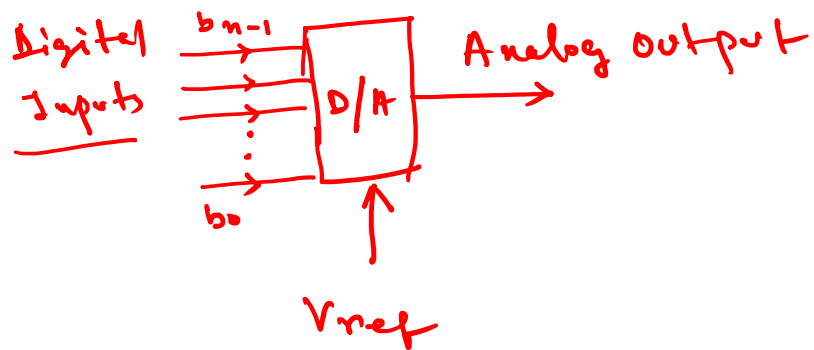
$$2^N \geq 100, \quad \underline{N = 7}$$

⊕ what will be the binary equivalent of 2.01V?

- How many bits are there in the system?
 - what is the resolution?
- } These information is required to answer the question

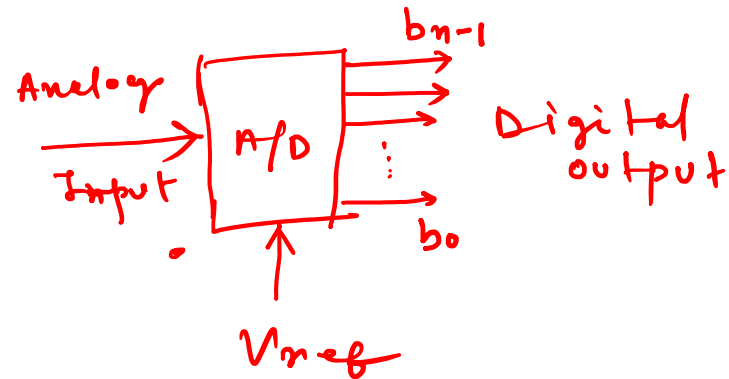
$$\text{Analog / Actual voltage} = \text{Resolution} \times [\text{Decimal eq. of binary bits}]$$

DAC



(Define Resolution)

ADC



(Define Resolution)

Digital to Analog Converter (DAC)

b_3	b_2	b_1	b_0	Decimal Eq.
0	0	0	0	0
0	0	0	1	1
0	0	1	0	2
0	0	1	1	3
.	.	.	.	.

Decimal Equivalent

$$= b_3 \times 2^3 + b_2 \times 2^2 + b_1 \times 2^1 + b_0 \times 2^0$$

$$= 16 \left[b_3 \cdot 2^{-1} + b_2 \cdot 2^{-2} + b_1 \cdot 2^{-3} + b_0 \cdot 2^{-4} \right]$$

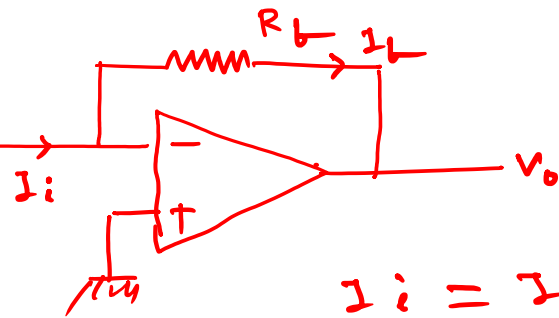
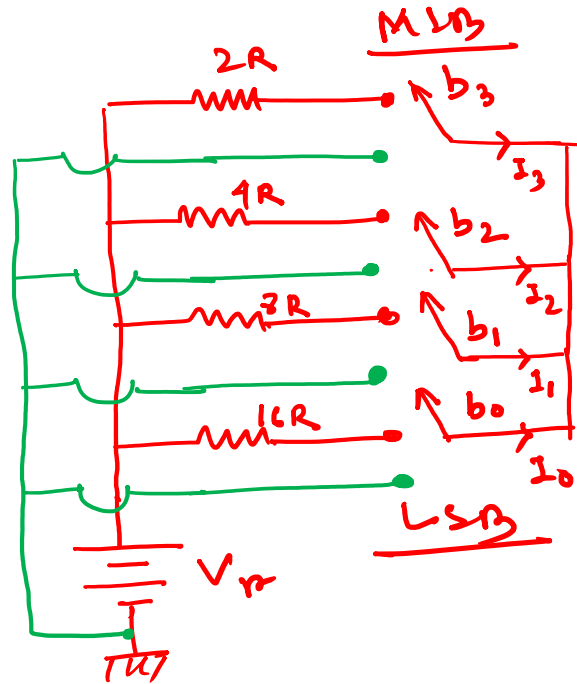
$$\checkmark = 16 \left[\underbrace{\frac{b_3}{2}}_{\rightarrow \text{MSB}} + \frac{b_2}{4} + \frac{b_1}{8} + \underbrace{\frac{b_0}{16}}_{\rightarrow \text{LSB}} \right]$$

$b_n \Rightarrow 0/1$ (Can be realized using switches)

Types of DAC

- Binary Weighted Resistor DAC
- Weighted Capacitor DAC
- R-2R Ladder DAC

Binary weighted Resistor type DAC



$$I_i = I_f$$

$$I_i = I_3 + I_2 + I_1 + I_0$$

$$V_o = -I_f \cdot R_f = -I_i R_f$$

$$= -R_f \left[\frac{V_r}{2R} \cdot b_3 + \frac{V_r}{4R} \cdot b_2 + \frac{V_r}{8R} b_1 + \frac{V_r}{16R} b_0 \right]$$

$b_n \Rightarrow 0/1$ (switch)

$$V_o = -\frac{V_r}{16} \left(\frac{R_f}{R} \right) \cdot 16 \left[\frac{b_3}{2} + \frac{b_2}{4} + \frac{b_1}{8} + \frac{b_0}{16} \right]$$

Resolution

Decimal Equivalent

$$= K \cdot V_{ref} \left[\frac{b_3}{2} + \frac{b_2}{4} + \frac{b_1}{8} + \frac{b_0}{16} \right]$$

Scale factor

$$= V_{FSR} \left[\frac{b_3}{2} + \frac{b_2}{4} + \frac{b_1}{8} + \frac{b_0}{16} \right], V_{FSR} = K V_{ref}$$

Limitation

- Large no. of resistors are required
- Resistor range is also large (Difficult for fab)
- Fabrication of highly precise resistors are difficult (Tolerance 0.5%)

- Current through b3 (MSB) resistor $I_3 = V_r/2R$
- Current through b2 resistor $I_2 = I_3/2$
- Current through b1 resistor $I_1 = I_2/2$
- Current through b0 resistor $I_0 = I_1/2$

$V_{FSR} = FSR \Rightarrow$ Full ~~and~~ scale Range

$V_{FSR} = K \cdot V_{ref} \Rightarrow$ Controls/Defines the resolution

$$\text{Resolution} = \Delta V = \text{LSB} = V_{\text{LSB}} = 1 \text{ LSB} = \frac{V_{FSR}}{2^N}$$

$\hookrightarrow$ basic step size

Full scale value = Full scale output = $V_{FSV} = V_{FSO} = (1 - 2^{-N}) V_{FSR}$

⑦ ~~Q~~ V_{FSV} is 1 LSB less than V_{FSR}
(ΔV)

Let $K=1$, $V_{ref}=1V$, $N=3$

$$\text{Resolution} = \Delta V = 1 \text{ LSB} = V_{\text{LSB}} = \frac{K \cdot V_{ref}}{2^3} = \frac{1}{8} V$$

$$000 \rightarrow 0V \quad V_{FSR} = K V_{ref} = 1V$$

$$001 \rightarrow 1/8 V \rightarrow V_{FSR}/8 \rightarrow V_{FSR}/2^N$$

$$010 \rightarrow 2/8 \rightarrow 2V_{FSR}/8 \rightarrow 2V_{FSR}/2^N$$

$$011 \rightarrow 3/8 \rightarrow 3V_{FSR}/8 \rightarrow 3V_{FSR}/2^N$$

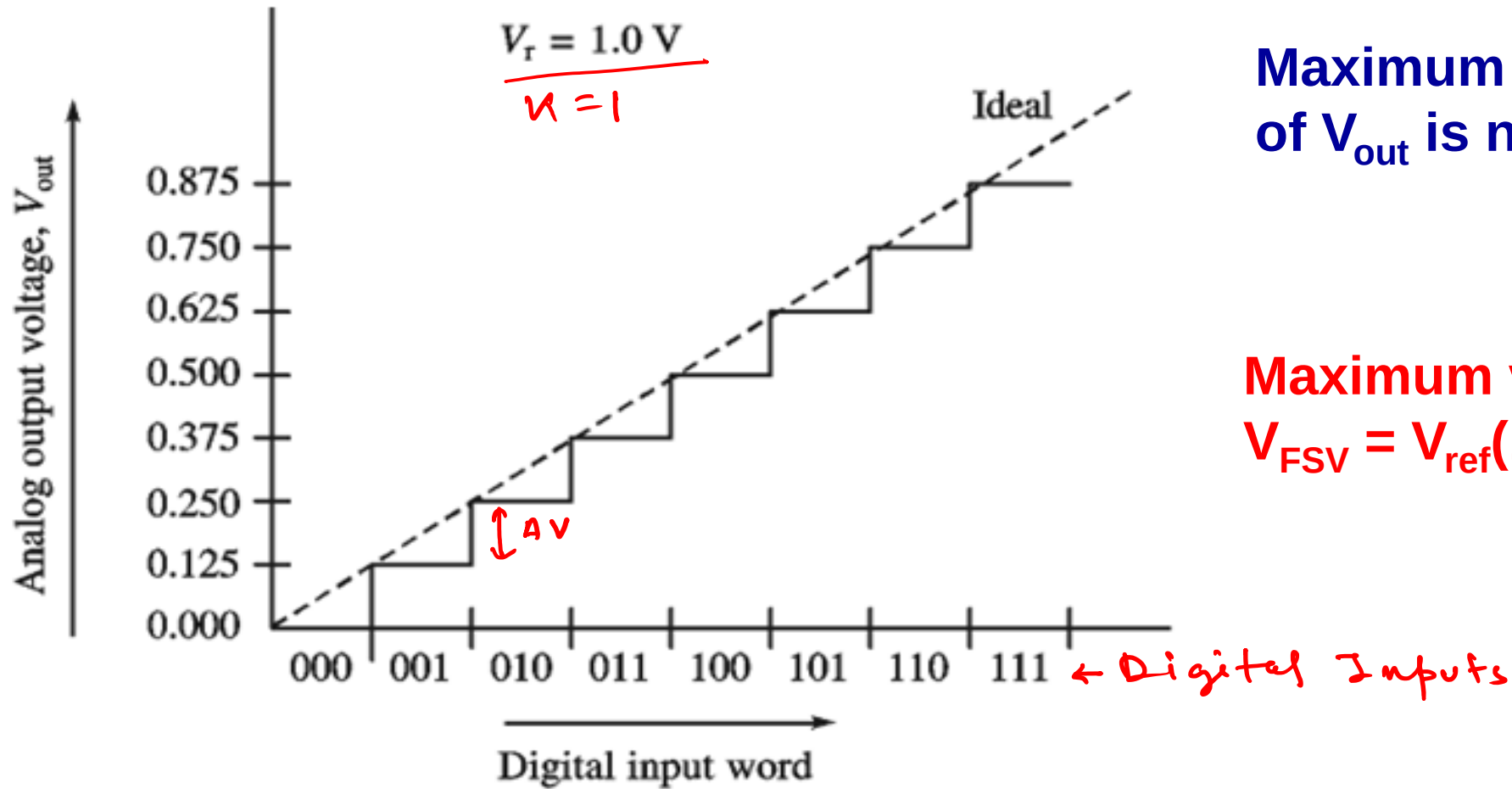
$$100 \rightarrow 4/8$$

$$101 \rightarrow 5/8$$

$$110 \rightarrow 6/8$$

$$111 \rightarrow 7/8 \rightarrow V_{FSV} \text{ or } V_{FSO} \left(\text{Output at the DAC when all the bits are '1'} \right)$$

3-bit DAC



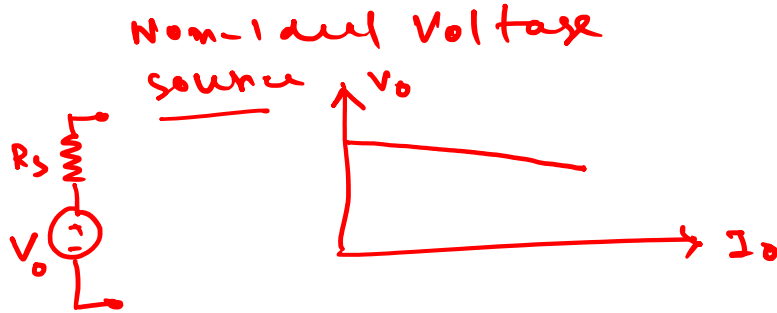
Maximum value (full scale) of V_{out} is not V_{ref} .

Maximum value (full scale)
 $V_{FSV} = V_{ref}(1-2^{-N})$ (or) $V_{ref} - V_{LSB}$.

Transfer characteristics of an ideal 3-bit D/A converter.

Limitations of Binary Weighted Resistor Type DAC

(1) **Reference Voltage Source should be Ideal:** The reference voltage source output must stay constant even when its output current is changing, i.e. its source resistance must be zero.

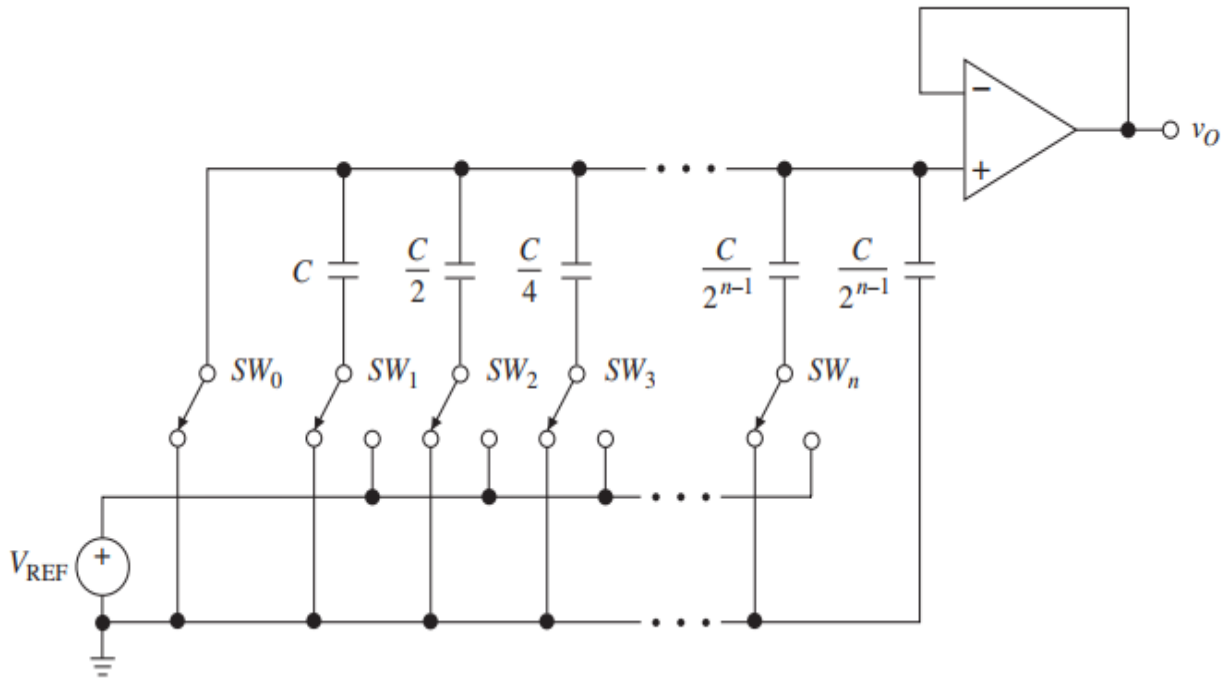


(2) **Wide Range of Precise Resistor Values:** Resistor values must be very accurate (Tolerance 0.5 %) which is difficult to fabricate. Range of values also increases with the no of bits. For Example: Resistances from 10 k Ω to 20.48 M Ω are required for 12-bit DAC

❖ Weighted Resistor DAC is suitable for N $\leq$ 6 (<6 bits)

Weighted-Capacitor DACs

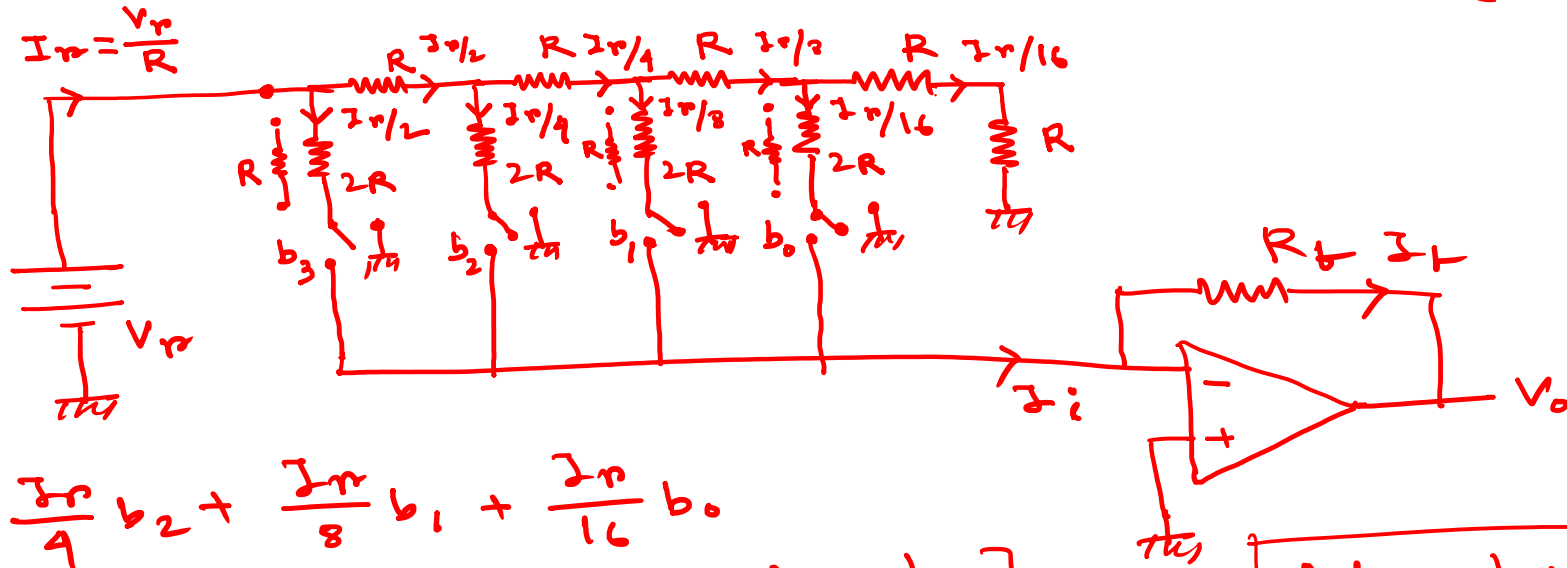
- Complex MOS ICs and microcomputers require on-chip DAC capabilities achievable using only MOSFETs and capacitors
- MOSFETs and MOSCAPs are easy to fabricate as they are natural components for the fabrication technology



- The circuit can be viewed as switched-capacitor counterpart of the weighted-resistor DAC
- Using elementary capacitor-divider principles, we find $v_O = V_{REF} C_r / C_t$, where C_r represents the sum of all capacitances connected to V_{REF} . C_t is the total capacitance of the array. $C_r = b_1 C + b_2 C/2 + \dots + b_n C/2^{n-1}$;
 $C_t = C + C/2 + \dots + C/2^{n-1} + C/2^{n-1} = 2C$.
Therefore, $v_O = V_{REF}(b_1 2^{-1} + b_2 2^{-2} + \dots + b_n 2^{-n})$
- Weighted Capacitor DAC is suitable for $N \leq 10$ (< 10 bits)

Inverting R-2R Ladder DAC

$b \Rightarrow$ switch (can be realized with BJT or MOSFET)
 $b \rightarrow 0/1 \rightarrow$ connected to the opamp
 $\quad \quad \quad \rightarrow$ connected to gnd



$$I_r = I_i$$

$$I_i = \frac{I_r}{2} \cdot b_3 + \frac{I_r}{4} b_2 + \frac{I_r}{8} b_1 + \frac{I_r}{16} b_0$$

$$\begin{aligned} \therefore V_o &= -I_i \cdot R_f = -I_r \cdot R_f \left[\frac{b_3}{2} + \frac{b_2}{4} + \frac{b_1}{8} + \frac{b_0}{16} \right] \\ &= -\frac{R_f}{R} V_r \left[\frac{b_3}{2} + \frac{b_2}{4} + \frac{b_1}{8} + \frac{b_0}{16} \right] \\ &= K V_r [B_{in}] \\ &= V_{FSR} \cdot [B_{in}] \end{aligned}$$

Advantage

The circuit is realized with only two different values (R and $2R$) of resistors

Current is divided by two at each node. This is the property of R-2R Ladder Network

1. Design a weighted resistor 8 bit DAC with a FS of 2.55 V

$$FS = V_{FS0} = V_{FSV} = 2.55, \quad K=1, \quad V_{ref}=?$$

$$V_{FSV} = K \cdot V_{ref} \cdot \frac{2^N - 1}{2^N} = 2.55, \quad \underline{V_{FSR} = K V_{ref}}$$

$$\begin{aligned} \therefore V_{ref} &= 2.55 \times \frac{2^N}{2^N - 1} \\ &= 2.55 \times \frac{256}{255} \\ &= \underline{2.56 \text{ V}} \end{aligned}$$

$$\text{Resolution} = \Delta V = \underline{0.01 \text{ V}}$$

2. The basic step of a 9 bit DAC is 10.3 mV. If 000000000 represents 0 V, what output is produced if input is 101101111?

$$\begin{aligned}\text{DAC output } V_o &= \text{Resolution} \times [\text{Decimal equivalent of the binary bits}] \\ &= 10.3 \text{ mV} \times 367 \\ &= \underline{3.7801 \text{ V}}\end{aligned}$$

$$\text{To find } V_{FSO} = \text{Resolution} \times [11111111]$$

$$\begin{aligned}V_{FSR} &= V_{FSO} + 1 \text{ LSB} \\ &= V_{FSO} + 4 \text{ V}\end{aligned}$$